Replacing flow dissector

State of Art

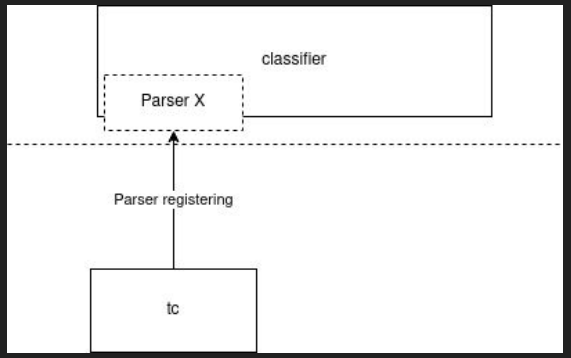
- Flow dissector is used by
  - Classifiers
    - Flower and others
  - (Some) Net drivers
  - eBPF hook (flow dissector program)
  - Etc...
- Static “big” parser
  - Supports many protocols and fine grained control over the flow keys
    - But… Not so good at handling tunneling https://lore.kernel.org/netdev/20210617161435.8853-1-vadym.kochan@plvision.eu/
  - What if I’m interested in just a 5 tuple parser?
    - Although we have control over the keys, there’s a performance overhead
      - https://netdevconf.info/0x15/session.html?Replacing-Flow-Dissector-with-PANDA-Parser

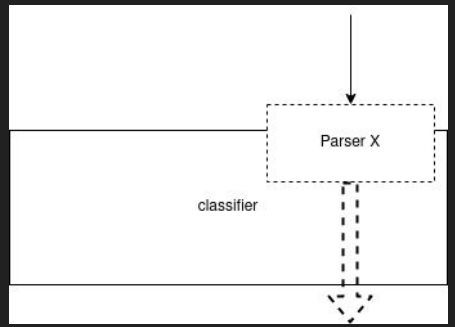
Parser as a dynamic component of a classifier

- Overcome flow dissector limitations
  - Better performance (Spoilers!)
  - Better tunneling handling
  - Better maintainability
    - Have you seen the flow dissector code recently?
  - Better flexibility
    - I only care about X tuples...

Architecture

- Register the parser to the classifier
  - Doable via an argument to the tc command line
- Associate the parser X to the filter Y
  - Once the parser is visible, this is trivially done
  - Store a struct ops to be invoked later
- Invoke parser X on classification
  - From the struct ops, invoke the parser entry point
- Inspirations drawn from act_ife
  - https://elixir.bootlin.com/linux/latest/source/net/sched/act_ife.c





Final thoughts

- If this got you interested and you want to know more…
  - Don’t miss Tom’s talk
  - Hit up Tom Herbert or Pedro Tammela
- Ultimate goal is programmable parsing hardware offloading!